Ibrahim Khalil

Storage & SAN | Backup & DR | Virtualization | Cloud | Kubernetes & DevOps

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PROFESSIONAL SUMMARY

Professional Certified Engineer & Technology Architect with 12+ years of experience designing, implementing, administering, migrating and supporting enterprise on-premises and cloud infrastructure. Key skills include:

- Enterprise Storage Dell EMC, Hitachi Vantara, NetApp, IBM, HPE Architecture, Implementation, Administration, Replication & Migration
- Commvault, NetBackup, Veeam, Acronis, EMC NetWorker, Avamar, DataDomain, IDPA, Backup/Recovery Implementation & Administration
- Cisco and Brocade SAN Fabric Implementation and Management
- Cloud-native serverless and Cloud-based services with OCI, AWS, GCP, Azure Cloud Technology
- Containerization Technology with Docker, Kubernetes & CI/CD pipeline tools using Jenkins, GitHub, AWS CodePipeline, Azure Pipelines
- Rack, Converged and Modular Chassis Server Implementation and Administration
- VMware vSphere, vCenter, vSAN, Microsoft Hyper-V, Proxmox VE and Data Center Virtualization
- Linux & Windows system administration and PowerShell, Bash & Python Scripting
- Data Center Infrastructure Management and Maintenance

PROFESSIONAL EXPERIENCE

Principal Engineer, Platform Infrastructure & DevOps — *July 2022 – Present*

Nagad Ltd. (Digital Financial Services), Dhaka, Bangladesh

Responsibilities:

- Implement, administer and optimize on-premises and multi-cloud infrastructure across **Servers**, enterprise **Storage**, **SAN** Fabric Switch, **Backup/Recovery**, **VMware/Hyper-V**, **Linux/Windows** and **Container** platforms.
- Deploy and optimize secure, scalable cloud services across **AWS**, **Azure**, **GCP** and **OCI**, including IAM/RBAC, compute, networking, storage, security policies and data protection.
- Implement **Docker** and **Kubernetes** platforms and **CI/CD** pipelines using **Jenkins**, **GitHub**, AWS **CodePipeline** & Azure Pipelines to automate application build, test and deployment workflows.
- Administer **Linux & Windows Servers, Clusters**, including **patching**, **vulnerability** assessment, **OS hardening** and advanced **troubleshooting**; monitor infrastructure with **Prometheus**, **Grafana**, **CloudWatch** and **ELK**.
- Lead geo-separated **storage replication** and **business-continuity** operations using Hitachi TrueCopy/SI/HUR/GAD, NetApp SnapMirror, SnapVault and Dell EMC RecoverPoint/SRDF; participate in periodic **DR drills**.
- Implement **backup and recovery** for business-critical applications, **VMs**, **Oracle**, **MySQL** and MS **SQL Server databases** using **Commvault**, **NetBackup** and **Veeam** in line with SLA, audit, regulatory compliance, restoration and DR requirements.
- **Data center** infrastructure management, monitoring, maintenance, **capacity planning**, **hardware lifecycle**, and operational support.

Lead Engineer, Server, Storage and Backup Administrator — *September 2020 – June 2022*

Nagad Ltd. (Digital Financial Services), Dhaka, Bangladesh

Responsibilities:

- Planned, deployed and administered heterogeneous **Storage & Backup** platforms integrated with Linux, Windows and VMware.
- Deployed and managed **GCP, AWS and Azure** services for secure, scalable and cost-effective cloud operations.
- Installed, configured & administered **Linux/Windows** systems, Clusters, including patching, vulnerability assessment and OS **hardening**.
- Administered virtualized environments using **VMware ESXi**, **vSphere HA, vSAN** and **Hyper-V Cluster** to ensure optimal **resource allocation**, conduct **capacity planning** and **performance tuning** to meet evolving business demands.
- Provided **L3 troubleshooting** and **performance optimization** for complex Server, Storage and Backup infrastructure incidents.
- **Performed storage provisioning**, **Capacity Planning**, Commissioning/Decommissioning, Cloning, Snapshots and Host mapping.
- Managed **SAN Fabric switch** and configured **VSAN**, **FC Zoning**, **ISL Trunking, Port Channel** & **IVR, NPV/NPIV, NX-OS, FOS Fw upgrade.**
- Operated centralized **Backup systems** using **Commvault** & **Veeam** to maintain data protection, retention and scheduling.
- Led maintenance, monitoring, firmware/patch lifecycle activities and OEM/vendor escalation for critical infrastructure issues.
- Health Checkup, Monitoring, Reporting and ensuring smooth operation of IT Infra, Server, Storage and Backup systems.

Asst. Manager, Enterprise Solution and Services — *July 2019 – August 2020*

Smart Technologies (BD) Ltd, Dhaka, Bangladesh

Responsibilities:

- **Led** the **Storage & Backup** technical team, resource onboarding and potential skill development.
- Delivered end-to-end **SAN/NAS Storage**, **Server**, **Archiving**, **Backup & Recovery**, **Converged Infrastructure**, and **VMware/Hyper-V** projects, from solution **design** and **implementation** through testing, migration, and production **go-live**.
- Installed, implemented, configured and provided **L3 support** for **Storage**, **Backup** & Recovery, **Cisco**/**Brocade** SAN, **VMware vSphere**/vSAN and **Hyper-V** solutions.
- Performed **Data Migration** using Storage & host-based migration technologies in **tech-refresh projects** with minimal downtime.
- Worked with OEM Professional Service team and customers on complex implementations, PoCs, project definitions, SoW, assessments, design reviews and technical specifications.

Asst. Manager, Customer Service and Implementation Engineer **(Server, Storage, Backup and Recovery)** ***July 2016 – June 2019***

Computer Services Limited (Dell EMC Authorized Partner), Dhaka, Bangladesh

Responsibilities:

- Provided onsite L3 support for production data centers, including P1 incidents, DU/DL cases and complex break-fix troubleshooting.
- Led **Storage, Server, SAN, Backup & Recovery project delivery**, coordinating implementation activities, milestone & customer requirements.
- Planned and executed **large-scale data migrations and technology-refresh projects** across heterogeneous OEM storage platforms, including migration planning, validation and production cutover activities.
- Implemented Dell EMC Celerra, CLARiiON, VNX/VNXe, VMAX/PowerMax, Unity, XtremIO and VPLEX integrated with Windows, Linux/Unix, VMware ESXi, Solaris, AIX and HP-UX
- Implemented RecoverPoint, Avamar, Networker, Data Domain and IDPA for centralized backup, recovery, replication and disaster recovery.
- Implemented Dell EMC Connectrix, Cisco and Brocade departmental/director-class SAN fabrics with Zoning, ISL Trunking & Port Channel.
- Performed storage provisioning, capacity planning, performance monitoring & lifecycle management across enterprise Storage, SAN Switch.
- Developed and implemented **Disaster Recovery (DR) plans** using RecoverPoint, Avamar, NetWorker and Data Domain, and performed periodic **DR drills and recovery validation** for FinTech, Banking/BFSI customers.
- Troubleshot backup/recovery issues and supported application/database protection for Oracle-DB, MySQL, MSSQL, MS Exchange workloads.
- Conducted **PoC, technical assessment and solution validation**, and contributed to **SoW** preparation for enterprise customers.

Customer Service Engineer **(Server, Storage, Backup and Recovery)** ***February 2014 - June 2016***

Computer Services Limited (Dell EMC Authorized Partner), Dhaka, Bangladesh

Responsibilities:

- Attended support calls to enterprise customers for EMC Storage, Backup Recovery & Server system solution and Services.
- Installed the Server, EMC Storage, Backup, SAN switch across customer production data centers using OEM best practices.
- hardware and Linux/Unix, Windows, VMware ESXi Operation system in multiple customer's Data Center with standard best practice.
- Performed storage provisioning, Windows, Linux & VMware ESXi Host Integration, Clone/Snapshot & Remote Replication configuration.
- Attended break-fix support call to FRU Replacement for EMC Storage and SAN Switch as OEM provided procedure.
- Configured Cisco/Brocade SAN switch, ISL and Zoning and provisioning hosts with various OS as like Windows, Linux/Unix and VMware ESXi based on customer requirements.
- Responsible for Firmware upgrade of EMC Storage, Backup and Cisco NX-OS & SAN Switch
- Provided 24x7 enterprise support services and coordinated escalated incidents with EMC Technical Support.

PROJECT SUMMARY

Successfully completed **50+ Projects** & **200+ implementations** across **Data Storage, SAN, Servers, Backup & Recovery, Disaster Recovery, VMware, Hyper-V, AWS, Azure, GCP, Kubernetes/K3s and CI/CD**, supporting **FinTech, Banking/BFSI, Telecom, Defense, Govt and Enterprise customers**. Selected noteworthy projects include:

- ✓ **Storage Tech Refresh & Data Migration** - Implemented Dell EMC Unity 500F and migrated CX4-480 workloads using SAN Copy, VMware vMotion and VPLEX Data Mobility.
- ✓ **VMware Data Center Consolidation** – Installed VMware ESXi, configured vCenter, vSphere HA, DRS, vMotion, vSAN, and consolidated physical workloads to improve availability, resource utilization and manageability.
- ✓ **Storage Technology Refresh, Backup & Data Center Migration** – Implemented Dell EMC XtremIO, Networker, Data Domain and RecoverPoint for enterprise storage, backup and disaster recovery.
- ✓ **Business Continuity & Disaster Recovery** – Implemented Dell EMC RecoverPoint and storage replication across DC / DR environments.
- ✓ **Data Center Relocation & Storage Migration** – Migrated Dell EMC CX4-480 workloads to VNX5400 and implemented RP for DR.
- ✓ **3-Site Active/Active Data Center** – Implemented Dell EMC VMAX 40K, Cisco MDS 9710 and VPLEX Metro for HA Active/Active DC.
- ✓ **Dell EMC Storage & Modular Server Infrastructure** – Installed and implemented Dell EMC SC5020F All-Flash Storage and PowerEdge MX7000 Modular Chassis Server infrastructure.
- ✓ **Enterprise NAS & File Services** – Implemented Dell EMC Isilon/PowerScale, Synology and QNAP NAS platforms for centralized file services, SMB/NFS, Active Directory integration, access permissions, snapshots and replication.
- ✓ **Commvault Enterprise Backup Solution** – Implemented Commvault backup infrastructure and disk-library integration with HPE StoreOnce and Hitachi VSP G350/VSP E790 storage across three data centers.
- ✓ **Veeam Backup & Replication** – Implemented VMware/Hyper-V backup infrastructure with HPE 3PAR repositories, backup proxies, retention policies, replication, Instant VM Recovery, file-level recovery and backup health monitoring.
- ✓ **Rubrik Data Protection** – Implemented policy-based backup and recovery for VMware and Hyper-V virtual infrastructure, including retention, recovery and backup validation.
- ✓ **Acronis Cyber Protect Cloud** – Implemented Hyper-V VM backup, M365 protection for Email and SharePoint, including backup and archiving.
- ✓ **Hitachi Active/Active Data Center** – Implemented Hitachi VSP 5100/VSP 5600/VSP G350, Cisco MDS 9706 SAN infrastructure and Hitachi Global-Active Device (GAD) & Universal Replicator (HUR) for Active/Active data center services.
- ✓ **NetApp AFF Enterprise Storage** – Implemented NetApp AFF A300 with ONTAP, SVMs, aggregates, volumes, LUNs, SMB/NFS, iSCSI/FC, SnapMirror and VMware integration.
- ✓ **HPE Storage & SAN Infrastructure** – Implemented HPE 3PAR/MSA storage and Brocade 6510 SAN switches, including CPGs, volumes, host mapping, Fibre Channel zoning and ISL configuration.
- ✓ **Cross-Platform VM Migration** – Migrated VMs between VMware and Hyper-V using StarWind V2V and VMware vCenter Converter.
- ✓ **Multi-Cloud Infrastructure** – Designed and administered infrastructure across AWS, Google Cloud Platform (GCP), Microsoft Azure and Oracle Cloud Infrastructure (OCI), covering compute, networking, storage, IAM/RBAC, security and data protection.
- ✓ **Kubernetes & Docker Container Platform** – Implemented containerized application platforms using Docker, Kubernetes and Rancher-managed K3s clusters for scalable and reliable application deployment.
- ✓ **CI/CD & Platform Automation** – Implemented Jenkins & GitHub Actions workflows for automated application build, test & deployment process.

TECHNOLOGY STACK / PLATFORM

Data Storage	▪ Hitachi VSP, G, E Series, HNAS Storage ▪ NetApp ONTAP AFF, FAS Series Storage ▪ HPE 3PAR, MSA SAN, Nimble, Alletra Storage ▪ EMC Symmetrix VMAX/PowerMax Series Storage ▪ Dell EMC PowerStore, PowerVault ME Storage	▪ Dell EMC Unity, Compellent SC Series, EqualLogic PS Series Array ▪ EMC CLARiiON, VNX Unified, VNXe, XtremIO, VPLEX ▪ Dell EMC Isilon/ PowerScale, TrueNAS, Synology, QNAP NAS ▪ Replication Tools: Dell EMC RecoverPoint, SRDF, Hitachi GAD, TC, SI, HUR, NetApp SnapMirror, SnapVault
Backup & Disaster Recovery	▪ Commvault Backup and Recovery Data Protection ▪ Veeam Backup & Replication, Cloud Connect, MSP ▪ Veritas NetBackup & Backup Exec ▪ Acronis Cyber Protect Cloud, MSP	▪ EMC NetWorker, Avamar, IDPA, DPS ▪ Dell EMC PowerProtect / Data Domain Backup Storage ▪ HPE StoreOnce Backup Storage ▪ Immutable Backup, S3 Object Storage, Dell Tape Library
SAN Connectivity	▪ Cisco MDS, Brocade SAN switches, SAN Management Tools: Cisco DCNM/NDFC, Brocade WebTools/SANnav ▪ FC/iSCSI, VSAN, Zoning, ISL trunking, Port Channel, IVR, Host Connectivity, PowerPath, HDLM, MPIO, DM Multipathing	
Servers & Compute	▪ Dell PowerEdge Rack and Modular Chassis Server ▪ Lenovo / IBM Power Series Server	▪ HPE ProLiant, Superdome and Synergy Frames Chassis Server ▪ Cisco UCS Server Family | Huawei FusionServer
Virtualization & HCI	▪ VMware vSphere, ESXi, vCenter, vMotion, vSAN ▪ MS Hyper-V, Failover Clustering, Hyper-V Replica	▪ Proxmox VE, KVM, Sangfor, Huawei DCS/FusionCompute ▪ Dell EMC VxRail HCI
Cloud & DevOps	▪ AWS, Azure, GCP, OCI, GKE, EKS, AKS ▪ CI/CD: Jenkins, GitHub, Azure / AWS CodePipeline	▪ Docker, Kubernetes (K8s), K3s, Rancher ▪ Helm, Storage Class, PV, PVC, CSI Drivers
OS & Services	▪ Linux, RedHat, CentOS, Ubuntu ▪ Windows Server 2012/2016/2019/2022	▪ Active Directory (AD), DNS, DHCP, GPO ▪ NGINX, BIND, NFS, Squid (Proxy), HAProxy etc.

PROFESSIONAL TRAINING

- ***Hitachi Ops Center Administration & Replication;*** Hitachi Data Systems – **Singapore**
- ***Managing Hitachi Universal Replicator (HUR);*** Hitachi Data Systems – **Singapore**
- ***Managing In-System Replication;*** Hitachi Data Systems – **Singapore**
- ***Operating and Managing Global-Active Device;*** Hitachi Data Systems – **Singapore**
- ***Dell EMC VNX Unified Storage Installation and Maintenance;*** EMC Professional Training, EMC Education Services – **Singapore**
- ***Cisco & Brocade SAN Management;*** EMC Professional Training, EMC Education Services – **Singapore**
- ***Dell EMC AVAMAR Management, Installation and Configuration;*** EMC Professional Training
- ***Dell EMC VMAX3 Installation and Maintenance;*** EMC Professional Training
- ***Dell EMC NetWorker Implementation and Management;*** EMC Professional Training
- ***Data Protection and Management;*** EMC Professional Training
- ***VPLEX Implementation;*** EMC Professional Training
- ***NetApp ONTAP 9.8 Data Protection Administration***
- ***NetApp ONTAP 9.8 Cluster Administration and Data Protection***
- ***DCMDS - Configuring Cisco MDS 9000 Series Switches***
- ***VMware vSphere: Install, Configure***
- ***VMware vSAN: Deploy and Manage***
- ***Commvault Professional Foundations***
- ***Commvault Professional Advanced***
- ***Cisco Certified Network Associate (CCNA) Exploration 4.0***

PROFESSIONAL CERTIFICATIONS

- ***Google Cloud Certified - Professional Cloud Architect Certificate***
- ***Certified Kubernetes Administrator (CKA)***
- ***Hitachi Vantara Qualified Professional - Global-Active Device management and operations***
- ***Hitachi Vantara Qualified Professional - Universal Replicator management***
- ***Microsoft Certified Azure Fundamentals***
- ***Red Hat Certified Engineer (RHCE)***
- ***Red Hat Certified System Administrator (RHCSA)***
- ***Commvault Certified Professional***
- ***VMware Certified Specialist - vSAN***
- ***VMware Certified Professional – Data Center Virtualization (VCP-DCV)***
- ***Specialist - Implementation Engineer, PowerEdge MX Modular Version 1.0***
- ***Specialist - Implementation Engineer, Dell EMC Unity Solutions Version 2.0***
- ***Specialist - Technology Architect, Midrange Storage Solutions***
- ***Associate - Information Storage and Management Version 3.0***

EDUCATION AND QUALIFICATIONS

2018 **M.Sc. in Computer Science**
CGPA 3.58 out of 4.00. Jahangirnagar University, Savar, Dhaka-1342, Bangladesh.

2013 **B.Sc. (Engg.) in Electronics and Communication Engineering (ECE)**
CGPA 2.76 out of 4.00. Khulna University, School of Science Engineering & Technology, Khulna-9208, Bangladesh

RESEARCH AND PUBLICATION

WDM Transmission for Free Space Optics under Different Atmospheric Conditions.
STM Journals, Trends in Opto-electro & Optical Communications, Vol4, Issue1, ISSN: 2231-0401, 2014.
Reference: *https://engineeringjournals.stmjournals.in/index.php/TOEOC/article/view/1829*